



Graphs in Machine Learning

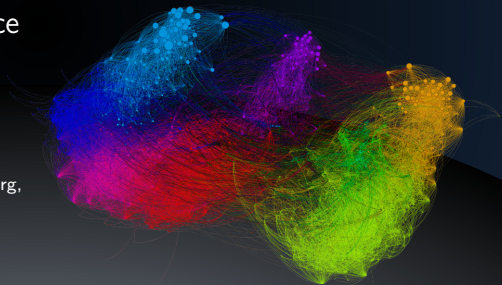
Resistive Networks

Physics and Effective Resistance

Michal Valko

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Partially based on material by: Ulrike von Luxburg,
Gary Miller, Doyle & Schnell, Daniel Spielman



Laplacians and Resistive Networks

How to compute the $\text{score}(v, m)$?

Idea₄: view edges as conductors

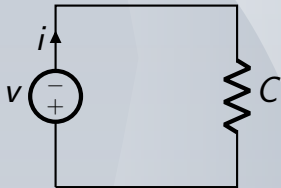
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$R \equiv \text{resistance}$

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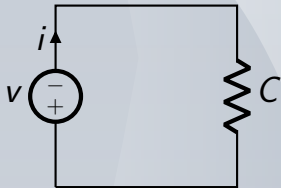
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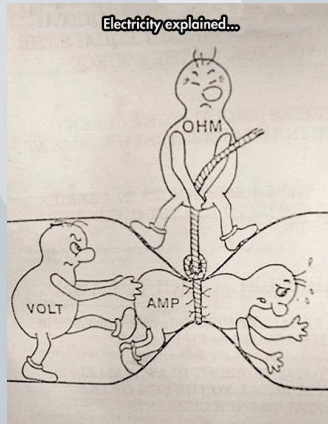
$R \equiv \text{resistance}$

$i \equiv \text{current}$

$V \equiv \text{voltage}$

$$C = \frac{1}{R} \quad i = CV = \frac{V}{R}$$

Resistive Networks: Some high-school physics



Source: Illustration commonly attributed to Vagonum

Resistive Networks

Resistive Networks

resistors **in series**

$$R = R_1 + \cdots + R_n \quad C = \frac{1}{\frac{1}{C_1} + \cdots + \frac{1}{C_N}} \quad i = \frac{V}{R}$$

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Effective Resistance on a graph

Take two nodes: $a \neq b$. Let V_{ab} be the voltage between them and i_{ab} the current between them. Define $R_{ab} = \frac{V_{ab}}{i_{ab}}$ and $C_{ab} = \frac{1}{R_{ab}}$.

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We treat the entire graph as a resistor!

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The effective resistance is a distance!

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<https://misovalko.github.io/mva-ml-graphs.html>